Arista Access Node Hardware/Environmental Fault Resolution Procedure for Whitefield POP

Node: WFD-ACC-31 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_8376

Description

The Arista access node, situated at the Whitefield POP, is prone to hardware/environmental faults due to its critical role in facilitating network access. ASIC memory parity errors, line card crashes, and power supply redundancy loss are indicative of such failures, often triggered by overheating, power surges, or hardware degradation. The Arista EOS operating system, underpinning the access node's functionality, relies on a complex interplay of protocols and mechanisms to ensure high availability and reliability. The hardware components, including line cards and power supplies, are designed with redundancy to mitigate single points of failure. However, when these components fail, the node's ability to forward traffic and maintain network stability is compromised, resulting in minor severity alarms like ALARM_8376. Understanding the underlying hardware and software architecture is crucial for effective troubleshooting and resolution of these faults.

Pre-requisite Checks

- Physically inspect the Whitefield POP for any signs of environmental stressors (e.g., water damage, extreme temperatures)

- Verify the status of redundant links to ensure no single point of failure

- Dispatch remote hands to the Whitefield POP for hands-on diagnosis and repair

Potential Root Causes

Cause	Probability
ASIC memory parity errors due to overheating	High
Line card crashes resulting from power supply redundancy loss	Medium
Hardware degradation from wear and tear	Low

Diagnostic Commands & Expected Outputs

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Command: show hardware forwarding asic, Expected Output: Detailed ASIC status, including temperature and error counts, Description: This command isolates issues within the ASIC, providing insights into potential overheating or hardware faults affecting forwarding capabilities.

Command: show redundancy, Expected Output: Status of redundant components, including power supplies and line cards, Description: This command assesses the redundancy of critical components, helping to identify single points of failure or compromised redundancy.
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```
Command: show environment, Expected Output: Environmental conditions, such as temperature and power supply status, Description: This command provides an overview of the node's environmental conditions, which can indicate if external factors are contributing to the fault.
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Resolution Steps

Step 1: Step: Replace Faulty Line Card, Command: hardware module replace , Description: If a line card is identified as faulty, replacing it ensures the restoration of full redundancy and prevents further failures. The specific module number should be identified through diagnostic commands.

Step 2: Step: Reset ASIC, Command: hardware forwarding asic reset, Description: In cases where ASIC memory parity errors are detected, a reset may clear transient faults. However, persistent issues may require hardware replacement.

Step 3: Step: Activate Redundant Power Supply, Command: no shutdown , Description: If a power supply redundancy loss is detected, activating the redundant supply ensures the node's power requirements are met, preventing further failures.

Post-Check Verification

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Command: show hardware forwarding asic, Expected Output: ASIC status indicating no errors or temperature issues, Description: Verifies that ASIC-related issues have been resolved, ensuring forwarding capabilities are stable.

Command: show redundancy, Expected Output: All redundant components are operational, Description: Confirms that redundancy has been restored for all critical components, reducing the risk of future failures.

Command: show environment, Expected Output: Environmental conditions within acceptable ranges, Description: Ensures that the node's environmental conditions are stable and not contributing to hardware faults.
```

Roll-back Procedures

- Step: Revert to Previous Line Card, Command: hardware module replace , Description: If the replacement line card introduces issues or does not resolve the fault, reverting to the previous module may be necessary to restore stability.

- Step: Disable Redundant Power Supply, Command: shutdown , Description: If activating a redundant power supply causes issues, it may need to be disabled to prevent further complications.

- Step: Revert ASIC Configuration, Command: hardware forwarding asic configuration revert, Description: In case an ASIC reset or configuration change exacerbates the issue, reverting to the previous configuration may help stabilize the node.